



DPHE Tree Plantation Tracker

বৃক্ষরোপণ ট্র্যাকার

INTERACTIVE USER GUIDE • UPDATED

Department of Public Health Engineering (DPHE), Bangladesh

Now including the Executive DSS analytics suite • Click any topic on the next slide to jump straight to it



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Introduction

A web-based tree plantation record-keeping, monitoring & decision-support system

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The DPHE Tree Plantation Tracker is used by DPHE offices nationwide to record, monitor, and report on trees planted at offices, compounds, and project sites — accessed entirely through a web browser, no installation required.

Each tree entry captures where it was planted, its species & category, caregiver, funding source & cost, current health status, and the maintenance activity carried out — feeding the Dashboard, Map View, six report types, and now the Executive DSS analytics suite.

NEW What's new in this edition: Clear Filter buttons and live filtered summary cards on the Monitoring & Records pages, a one-tap Google Maps "Get Directions" button per tree, and an entirely new Executive DSS module covering KPIs, risk, ranking, climate, budget, forecasting, and an interactive GIS map.

HQ (Head Office)

National level



Circle Office

e.g. Dhaka Circle



District Office

e.g. Gazipur, Feni



Upazilla / Pourashava Office

Where trees are planted & cared for



Login & User Roles

Sign in with your email & password — your role decides what you see

Role	Data Visibility (Scope)	Typical Permissions
HQ Admin	All circles, districts & upazillas nationwide	Full access — add/edit/delete, manage users, all reports, Executive DSS, DSS Configuration
Divisional (Circle Officer)	Own Circle (all districts/upazillas within it)	Add, monitor, view records/map, reports & export
District	Own District (all upazillas within it)	Add, monitor, view records/map, reports & export
Upazilla	Own Upazilla / Pourashava only	Add, monitor, view records, reports & export



Only an HQ Admin can add/remove users or delete records — unless individually granted that permission.



The Executive DSS is now a per-user permission too: HQ Admin always has it; Circle/District/Upazilla users only see the "Executive DSS" menu item once an HQ Admin checks "Executive DSS দেখা" for their account in User Management. DSS Configuration (editing the assumptions) always stays HQ-Admin-only.



Navigating the App

The main menu — what you see depends on your role & permissions

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Dashboard

Quick overview of plantation activity



Add Tree Record

Log a newly planted tree



Periodic Monitoring

Record follow-up visits



All Records

Search, filter, edit, delete



Map View

See planted trees by Circle



Plant Species

Master species & category list



Reports

Generate & export 6 report types



User Management

Create & manage accounts



Executive DSS

National/scoped KPIs, risk & forecasts



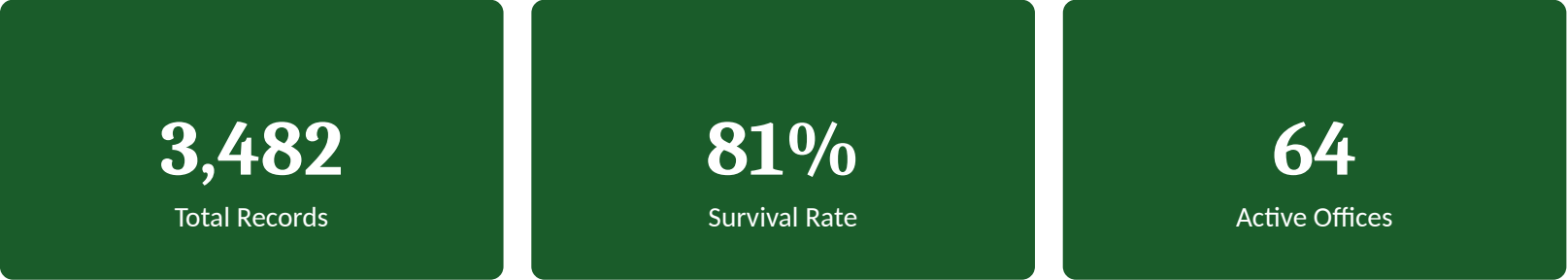
DSS Configuration

HQ Admin — edit every DSS assumption

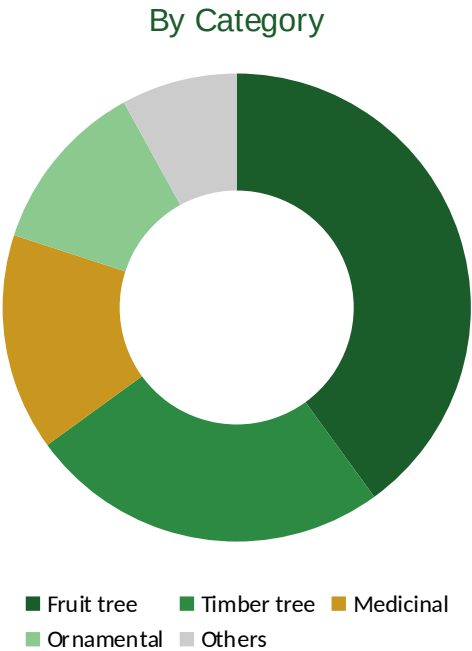
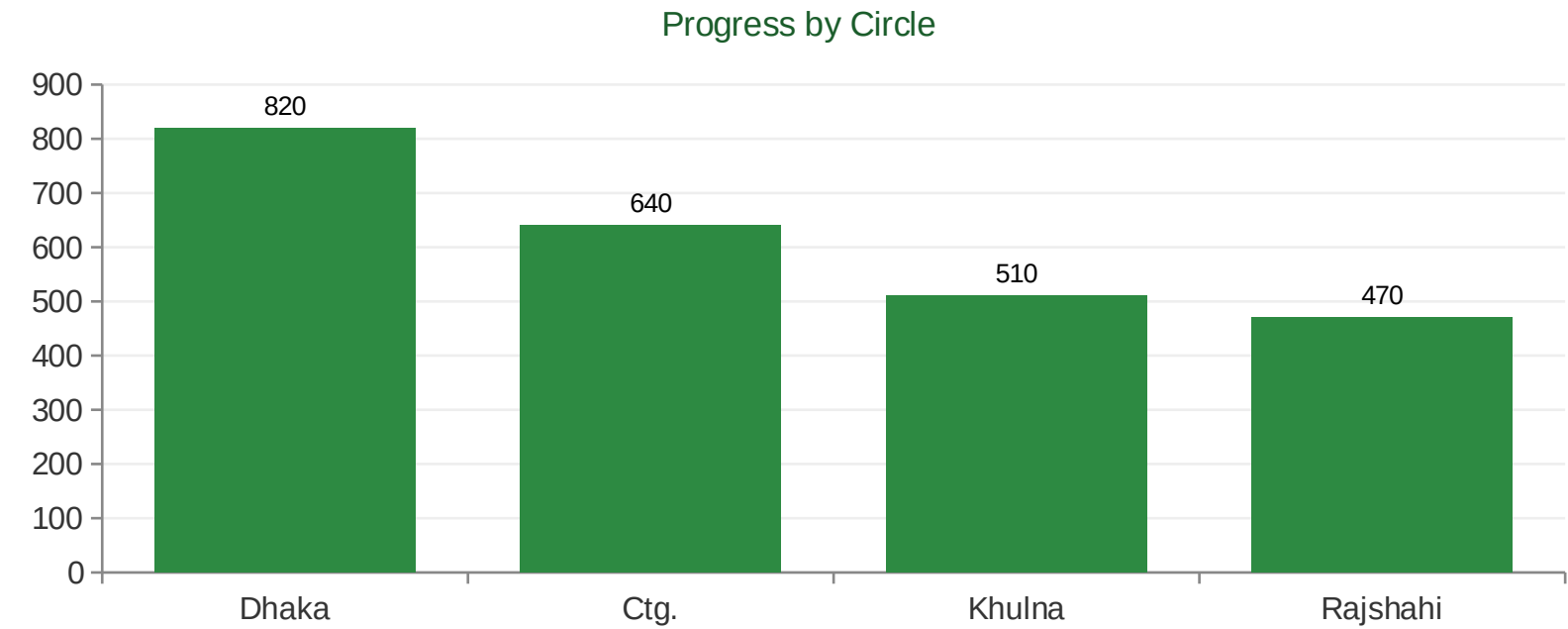


Dashboard

An at-a-glance summary of plantation activity in your scope



Also shown: Progress-by-Circle bar chart, Monthly Trend chart, and a Recent Records table.





Add Tree Plantation Record

Logged every time a new tree (or batch) is planted — * = required

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Location & Site

- Planting Level (HQ / Circle / District / Upazilla)
- Circle → District → Upazilla (cascading) *
- Site Name *
- GPS Coordinates — Auto Fetch or manual entry

Tree Details

- Tree Species * (Category auto-fills)
- Date of Plantation *
- Caregiver Name & Contact
- Funding Source & Cost (BDT)
- Initial Status (Alive / Care Required / Dead / Pending)

Photo Upload

Upload photos or capture directly from a camera on mobile.
Photos are auto-resized to ≤100 KB before saving.

Saving & Editing

- Click “Save Record” to submit a new entry
- Open a record from All Records and click Edit to correct it later — the form re-opens pre-filled



Care/Maintenance Activity

A controlled dropdown that keeps large reports short & readable

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Office Management



Fencing



By Caregiver

1

Others

Selecting “Others” reveals a free-text box to describe the activity in your own words.

Detailed Record Report: shows the full detail you typed, including any “Others: ...” text.

Office Register & Detailed Species Reports: group everything into just these 4 labels with counts — e.g. “Office Management: 320, Fencing: 210, Others: 45” — so a report stays readable even with thousands of entries.



Monitoring, Records & Map

Track visits, manage the full record list, and see plantations geographically

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Periodic Monitoring

- Filter by Circle or search by ID/site/species
- GPS Radius search finds nearby trees on-site
- Log visit date, updated status & notes
-  Clear Filter button resets every filter instantly
-  Live summary cards: Total Tree, Alive, Total Monitoring, Last Monitoring Date
-   Get Directions button per row — opens Google Maps directions from your current location straight to that tree's GPS point

All Plantation Records

- Filter by Circle, Category, or Status
- Search by ID, site, species, or caregiver
- View full details, edit, or delete a record
-  Clear Filter button resets every filter instantly
-  Live summary cards: Total Tree, Total Cost, Alive, Dead

Map View

- Trees plotted geographically, grouped by Circle
- Uses GPS captured at record creation
- Spot plantation coverage & gaps at a glance

 **NEW**

Both filter bars now update their summary cards live — filter by anything above and the Total/Alive/Cost/Dead (or Monitoring) numbers recalculate immediately, with no page reload.



Plant Species

The master list behind the Tree Species dropdown

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Every species carries a fixed Category, applied automatically wherever it's selected on the Add Tree Record form. Search by English or Bangla name to find a species quickly.



Fruit tree

e.g. Mango, Orange, Jamun



Timber tree

e.g. Mahogany, Segun



Medicinal

e.g. Neem



Ornamental

e.g. Plumeria



Others

Anything outside the above



Reports Overview

Build once, export three ways

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Report Builder

- Choose a Report Type (6 available — next slide)
- Set a From / To date range
- Set Scope — HQ Admins pick any Circle/District/Upazilla; other roles can only narrow within their own jurisdiction
- Click “Generate Report” to build it on screen



Print

Directly from the browser



CSV

Opens in Excel / Google Sheets



PDF

A formatted, shareable document



The Six Report Types

Each scoped and filtered the same way

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Summary Report

Totals, survival rate, cost & an office-wise breakdown



Detailed Record Report

Every individual tree record, row by row



Office Register Report

One row per office, in HQ→Circle→District→Upazilla order



Monthly Trend Report

Trees planted & cost over the last 12 months



Detailed Species Report

Species grouped by category, office by office



Species Report

Category & Survival Rate, most-planted first



Office Register Report — In Detail

Matches the traditional paper register format

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Name of the Office

Taken from the Office field set in User Management

Tree Count

Total trees recorded for that office

Species Category

Counts per category, e.g. "Fruit tree: 150"

Offices are listed by administrative hierarchy — not alphabetically:

HQ

Circle Office

District Office

Upazilla Office

...then the next Circle, and so on.



Because categories are shown as counts, this report stays compact even when a single office logs thousands of tree entries.



Executive DSS

Government-grade decision support analytics, built entirely on your plantation data

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What it is

A dedicated analytics layer above the operational pages — turns raw plantation records into KPIs, risk flags, rankings, climate accounting, budget ROI, forecasts, and an interactive risk/heat map.

Who sees it

Available to every role. HQ Admin always has it; Circle/District/Upazilla users need the “ Executive DSS” permission granted individually in User Management. Data stays scoped to their own jurisdiction, like every other page.

Filter bar & Presentation Mode

Pick a Circle → District → Upazilla scope (or leave national) — every module recalculates instantly. A one-click Presentation Mode hides editing controls and enlarges KPI cards for boardroom use.

From records to decisions

 **Plantation Records**



 **KPIs • EPI • Risk • Ranking**



 **Climate • Budget • Forecast • GIS**



 **Executive Decisions**



Nothing is hardcoded — every weight, threshold, cost, and rate the DSS uses lives in one place: DSS Configuration (HQ Admin only). Change a value there and every module & report reads the new figure immediately.



Executive KPIs & Performance Index

Fourteen headline numbers, plus one weighted score

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Executive KPI cards

- Total / Alive / Dead Trees
- Survival % — $\text{Alive} \div \text{Total}$
- Monitoring Completion — % with ≥ 1 visit logged
- Trees Requiring Care (Care Required status)
- Budget Used & Cost per Tree
- Budget Remaining — HQ Admin only, vs national allocation
- Best / Worst Circle, Best District — by Survival %
- Highest-Mortality District
- Monthly & Annual Growth — plantings vs previous month/year, anchored to the latest plantation date in the data

Environmental Performance Index (EPI)

$$\text{Score} = \text{Survival\%} \times w_1 + \text{Monitoring\%} \times w_2 + \text{Backlog Health\%} \times w_3$$

Backlog Health% = 100% – share of trees flagged “Care Required”. Weights (w_1 , w_2 , w_3) are configurable — default 0.5 / 0.3 / 0.2 — and always sum to how much each factor moves the 0–100 score.

Excellent

Good

Moderate

Critical

Score bands, each with a configurable minimum cutoff (DSS Configuration → A).



Shown as a half-donut gauge on screen, colored to match its band.
Renamed from “National Health Index” — this DSS measures tree plantation & climate outcomes, not human health.



Risk, Alerts & Recommendations

The DSS watches every circle for trouble — and tells you what to do about it

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Smart Alerts

- Overdue Monitoring — Critical/High/Medium by % of trees unchecked past the configured day threshold
- High Mortality — one or more circles above the mortality threshold
- Missing Data — records with no GPS or no photograph
- Budget Anomaly — over-allocated, remaining-budget low, or utilization high
- Care Backlog — Care Required count over threshold
- Sorted Critical → High → Medium → Low



Risk Assessment by Circle

- 6 checks per circle: High Mortality, Poor Monitoring, Care Backlog, Missing GPS, Missing Photograph, Overdue Monitoring
- Each triggered risk shows a suggested action
- Risk Score = number of risks triggered
- Priority: High (≥ 4) · Medium (≥ 2) · Low (≥ 1) · None



Executive Recommendations

- Auto-generated & prioritized from circle risks, bottom-performing districts, budget alerts, and the care backlog
- Each item shows its Reason and Expected Impact
- Top 12 kept, most urgent first — a focused list, not exhaustive



Performance Ranking

Circles, districts, and upazillas — ranked on one comparable score

Overall Score = Survival% × w_survival + Monitoring% × w_monitoring + Cost-Efficiency% × w_cost

Cost-Efficiency% rewards a lower cost per tree (cheaper per tree = higher efficiency). Weights are configurable in DSS Configuration.

Circle level

All Circles always shown in full — only 10 nationally, so no Top/Bottom split is needed.

District & Upazilla level

Split into Top 10 and Bottom 10 by Overall Score.

Minimum Sample Size

Any group below the configured minimum sample size is excluded — so a tiny sample can never rank as “best” or “worst.”

Rank	Name	Total	Survival%	Monitoring%	Cost/Tree	Overall Score
1	Dhaka Circle	820	86%	74%	₳412	81.4
2	Chattogram Circle	640	79%	68%	₳455	74.9



Climate Intelligence

Rule-of-thumb carbon accounting for the trees your teams have planted

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Carbon Stored (to date)

Age of each living tree $\times$ CO₂/tree/year $\times$ carbon fraction, summed

Annual Carbon Sequestration

Alive Trees $\times$ CO₂ Sequestered per Tree per Year $\times$ carbon fraction

Lifetime Carbon Storage

Annual Carbon $\times$ Assumed Tree Lifetime (years)

CO₂ Removed (to date)

Same age-weighted method, before the carbon-fraction step

Projected CO₂ Removal

Configurable horizons (e.g. 5 / 10 / 20 yr), with a stated confidence %

Greenhouse Gas Reduction

Annual CO₂ $\times$ configurable reduction multiplier

Oxygen Production

Alive Trees $\times$ O₂ produced per tree per year

PM2.5 Reduction / Air Quality

Alive Trees $\times$ PM2.5 filtered per tree per year (same underlying figure, two labels)

Carbon Lost — Dead Trees

Dead Trees $\times$ full assumed-lifetime carbon potential foregone



Standard rule-of-thumb estimates, not a species-specific or third-party-verified audit. Only living trees count toward sequestration. Every constant is adjustable in DSS Configuration → C.



Budget Intelligence & Predictive Analytics

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What we've spent, what it bought — and what comes next



Budget Intelligence

- Budget Spent — sum of recorded cost
- Budget Remaining — HQ Admin only, vs national allocation
- Cost per Alive Tree · Cost per Dead Tree
- Budget Efficiency % = Survival %
- Estimated Annual ROI (HQ Admin only) = this year's CO₂ value (Annual CO₂ × Social Cost of Carbon) ÷ Budget Spent — tiered Excellent / Acceptable / Low



Predictive Analytics

- A transparent projection, not a trained model
- Survival after 5 / 10 years = Alive × (1 – attrition rate)^{years}
- Projected Deaths (next year) = Alive – 1-year projected alive
- Maintenance Demand (next yr) = projected alive × per-tree maintenance cost
- Replacement Cost (next yr) = projected deaths × avg. cost/tree
- Budget Requirement (next yr) = Maintenance + Replacement



ROI values only the climate benefit — not shade, biodiversity, soil, or social value — so treat it as a conservative lower bound, never a complete economic appraisal. Carbon price & ROI tiers are editable in DSS Configuration → B.



The attrition rate (and every other rate above) is a single configurable assumption — edit it once on DSS Configuration and every forecast updates.



Scenario Simulator & GIS Intelligence Map

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"What-if" planning and a live, filterable risk/heat map



Scenario Simulator

- A side-by-side "what-if" comparison against the real Executive KPIs — entirely hypothetical and read-only
- Quick presets: target Survival %, +Maintenance %, +Monitoring %
- Manual entry for exact figures: budget, attrition rate, carbon price, CO₂/tree, monitoring %, forecast years, risk multiplier
- Manual values always override the matching preset
- Never edits, clones, or writes back to real records — reset anytime



GIS Intelligence Map

- Four heat-map layers: Carbon Density, Risk, Survival/Mortality, Budget
- Colored bands with sharp transitions exactly at each configured cutoff
- A live legend/scale always matches the current DSS Configuration → F thresholds
- Switch layers instantly — the map re-renders from the same filtered record set as every other module



DSS Configuration

Every DSS assumption and threshold, in one HQ-Admin-only place

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A. Performance Score Bands

B. Budget Allocation & Thresholds

C. Carbon & Climate Configuration

D. Tree Survival / Attrition

E. Monitoring Thresholds

Risk Assessment & Smart Alerts

Performance Ranking & EPI Weights

F. GIS Intelligence — Color Bands

G. Climate Intelligence — Forecasting



A schema-driven form — adding a new configurable value anywhere in the DSS means adding one entry here, not hand-writing another input. Changing a value instantly changes every DSS module & report that reads it — nothing is hardcoded.



Editing is HQ-Admin-only (separately gated from the “view Executive DSS” permission). A “Reset to defaults” option is available, and every field can carry a source reference/citation. Values sync via the shared backend, so one Admin's change appears for everyone.



User Management

Available to HQ Admins (or anyone granted the “users” permission)

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Creating a User

- Name, Username/Email, Password
- Role — HQ Admin, Divisional, District, or Upazilla
- Office — name & location, e.g. “Sadar Upazilla DPHE Office, Feni”
- Circle / District / Upazilla — sets their data visibility



The Office name entered here is exactly what appears as “Name of the Office” in the Office Register & Species Reports — enter it clearly and consistently.

Field-level Permissions

- Dashboard · Add Tree Record
- Monitoring · All Records
- Reports · Users
-   Executive DSS — view only; DSS Configuration editing always stays HQ-Admin-only



e.g. allow a District user to delete records, hide the Cost field from a role, or turn on Executive DSS for one Circle Officer without granting it app-wide.



Data Sync & Troubleshooting

How saving works, and quick fixes for common issues

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✓ Updated/Saved in Google Sheets

Your change synced successfully to the shared backend.

⚠ Sync failed

Only saved on this device. Check your connection and save again, or other users won't see it.

Quick Fixes

- Empty District/Upazilla dropdown? Select its parent field first.
- GPS auto-fetch failing? Allow location access & try outdoors.
- Report looks empty? Check the date range & scope filters.
- PDF Bangla text broken? Check your connection — the font loads once per session.
- NEW Get Directions button missing? That tree has no GPS coordinates recorded yet.
- Anything else — contact your HQ Admin.



DSS Configuration now syncs through the same shared backend as everything else (its own DSS_Config sheet): an HQ Admin's change appears for every user after their next refresh — no more per-device drift. Only HQ Admin can write it; other roles read it live (or don't see it at all without the Executive DSS permission).



Thank You

Grow every plantation record into a healthier, greener Bangladesh

[!\[\]\(3d8c13c92b853674f749aac6fa869926_img.jpg\) BACK TO MENU](#)

This guide covers the full app, including the Executive DSS analytics suite. For questions about the system, contact your HQ Admin.